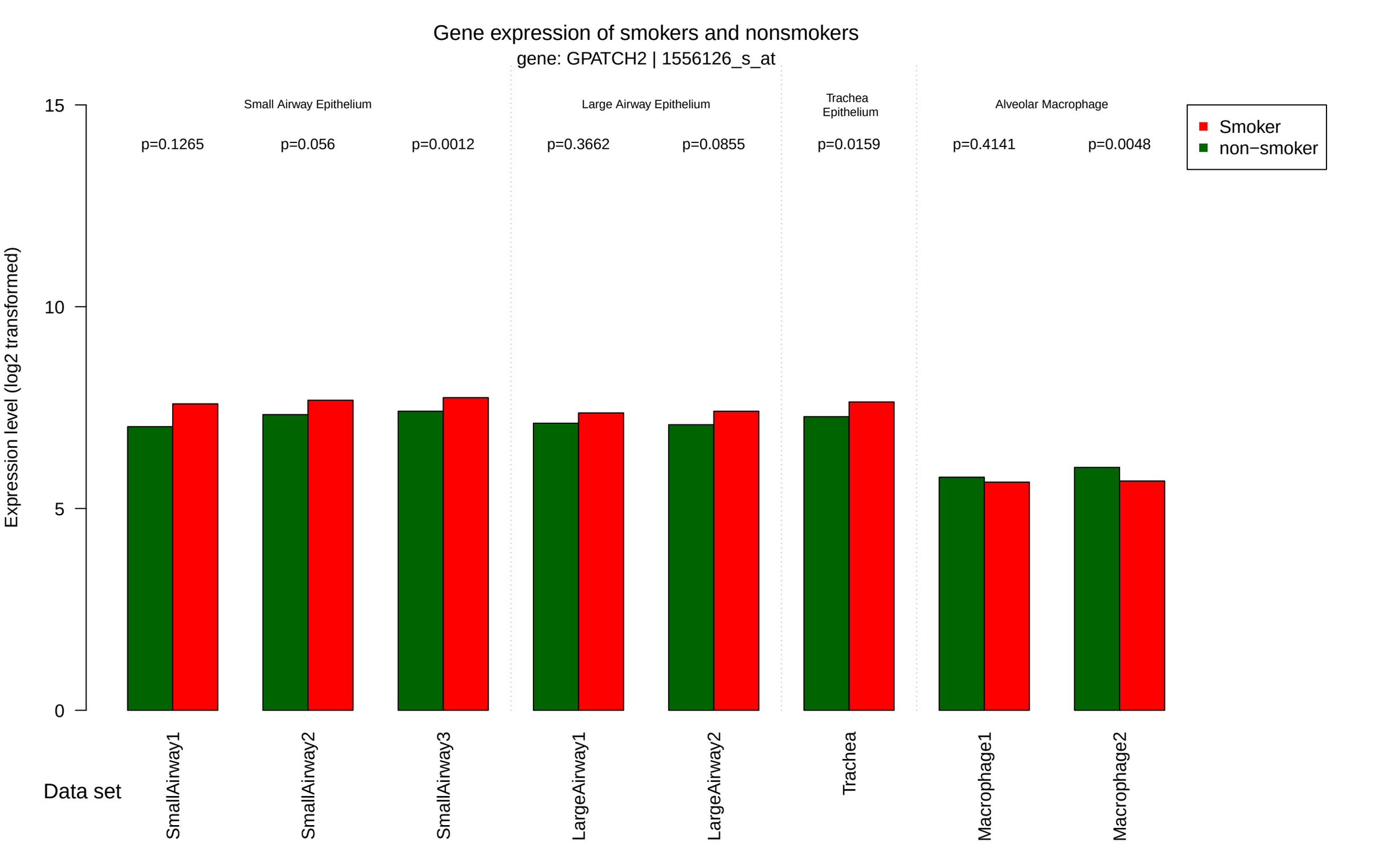
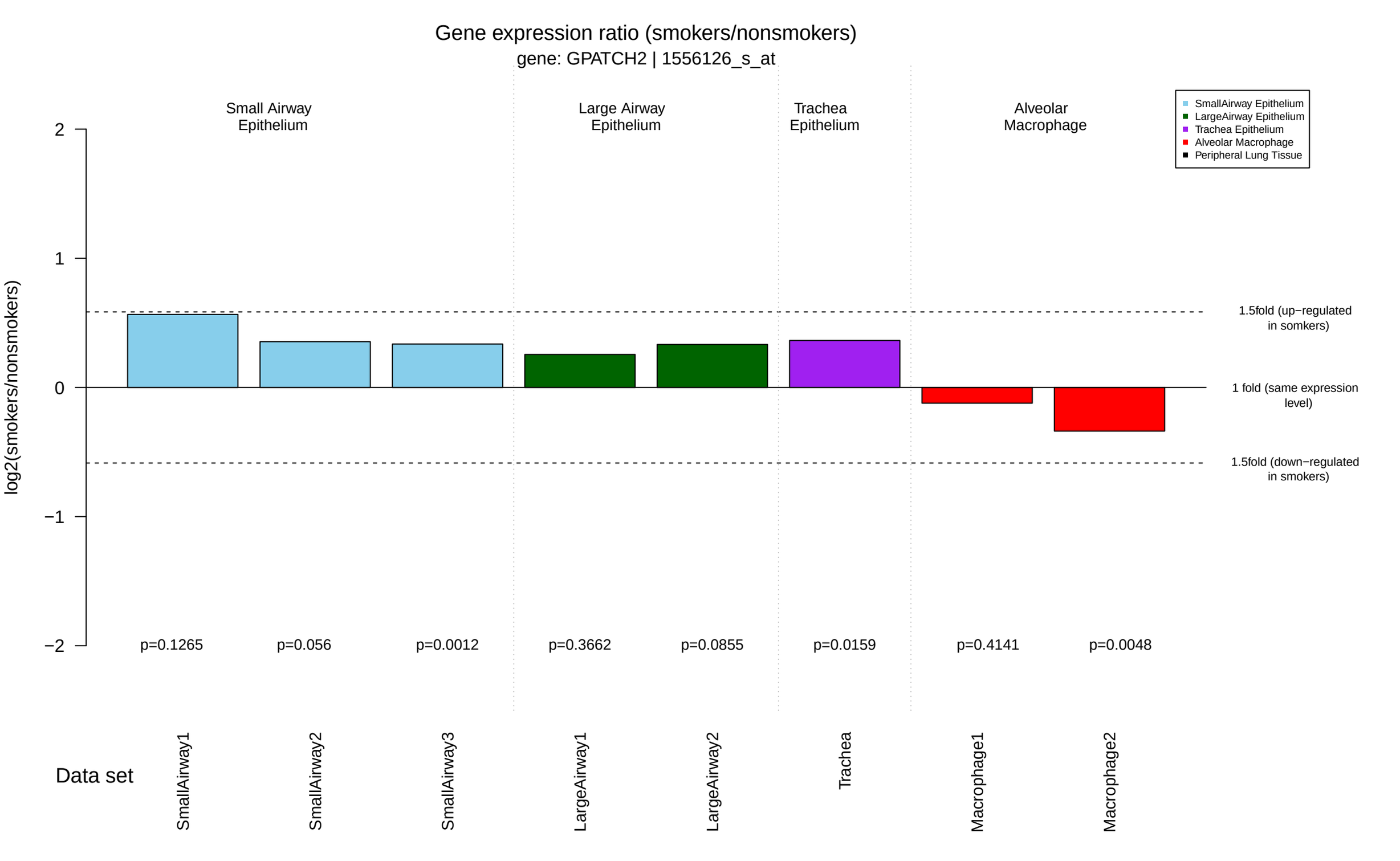
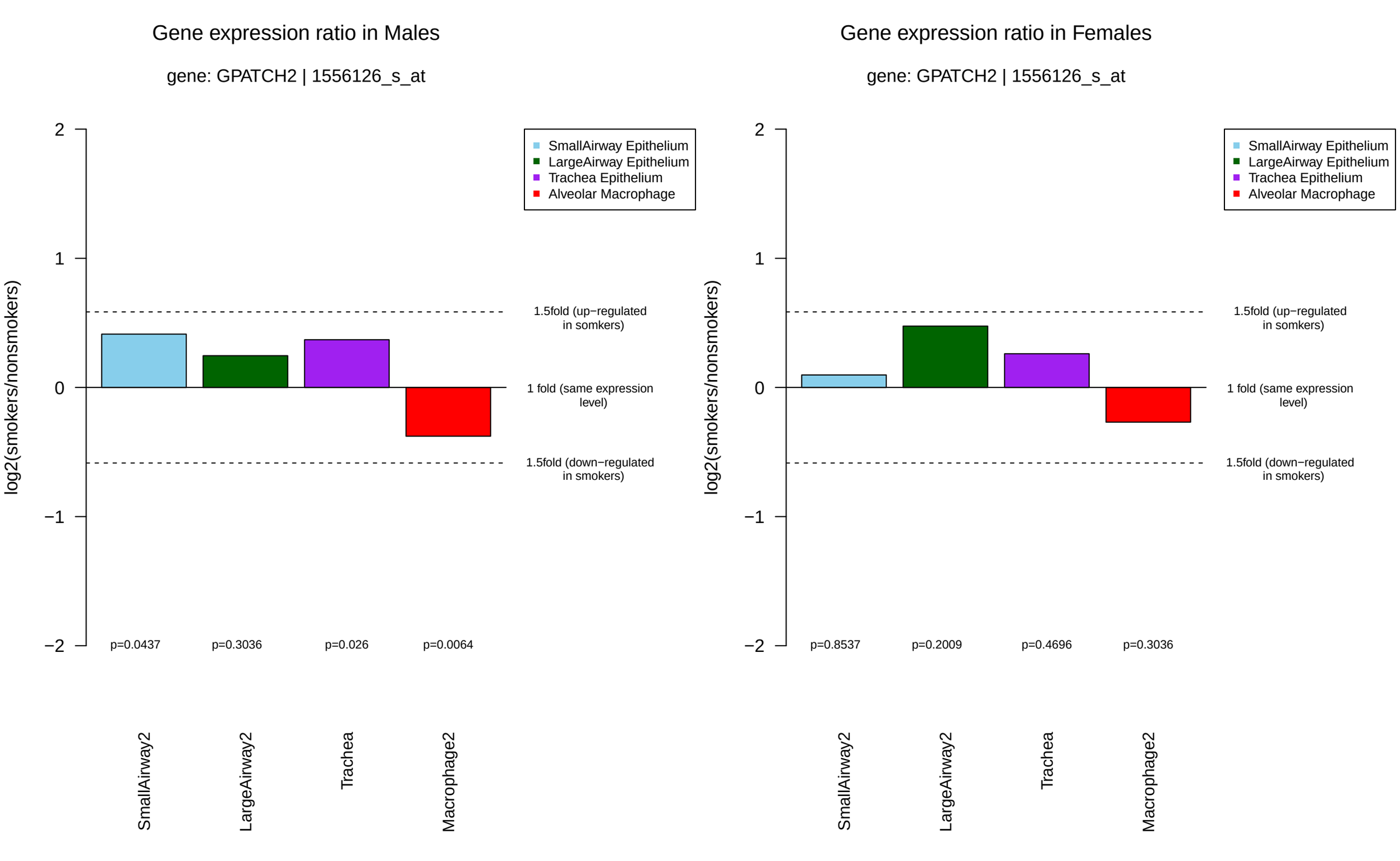
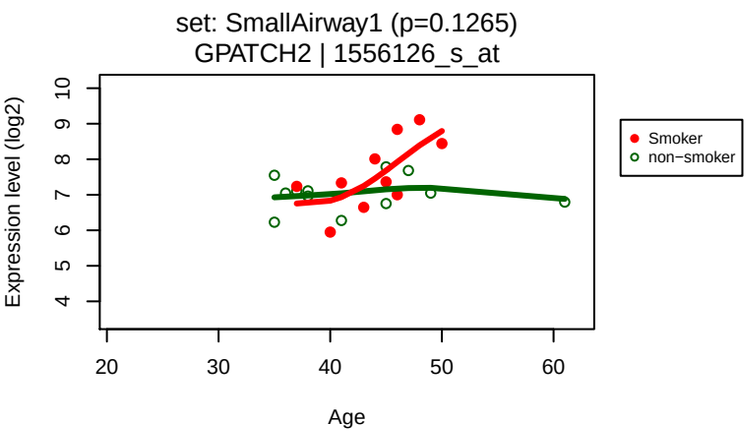
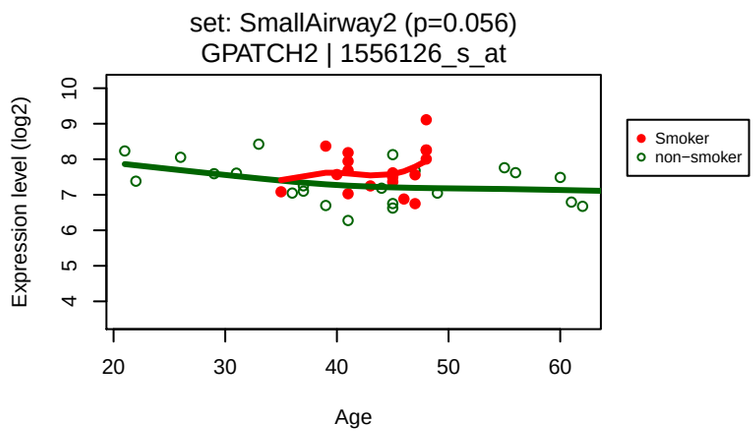
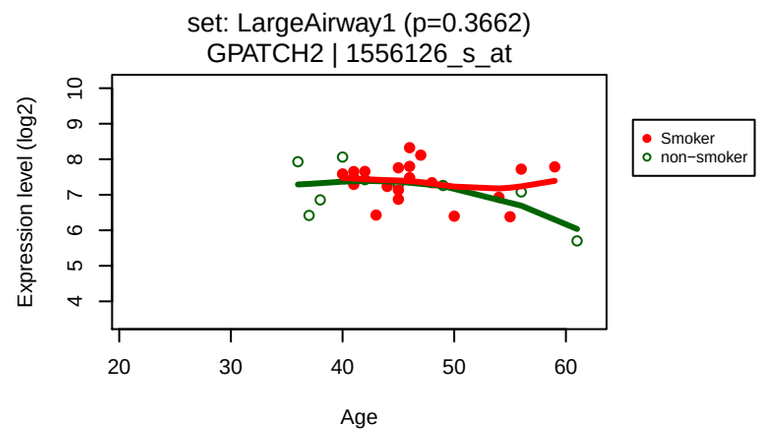

Gene expression of smokers and nonsmokers
gene: GPATCH2 | 1556126_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1265
p=0.056
p=0.0012
p=0.3662
p=0.0855
p=0.0159
p=0.4141
p=0.0048
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

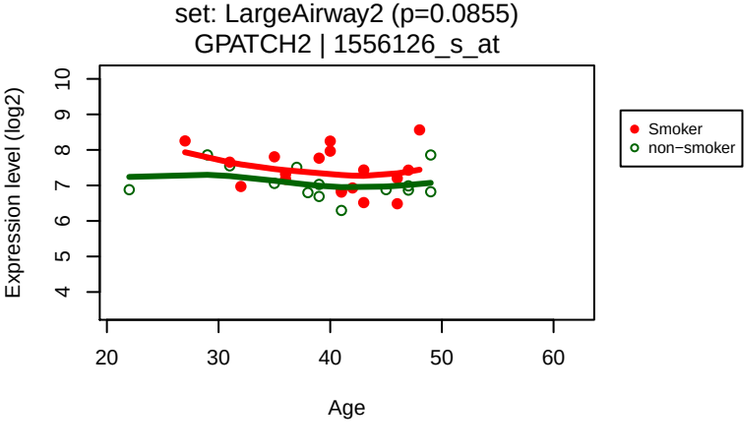

Gene expression ratio (smokers/nonsmokers)
gene: GPATCH2 | 1556126_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1265
p=0.056
p=0.0012
p=0.3662
p=0.0855
p=0.0159
p=0.4141
p=0.0048
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

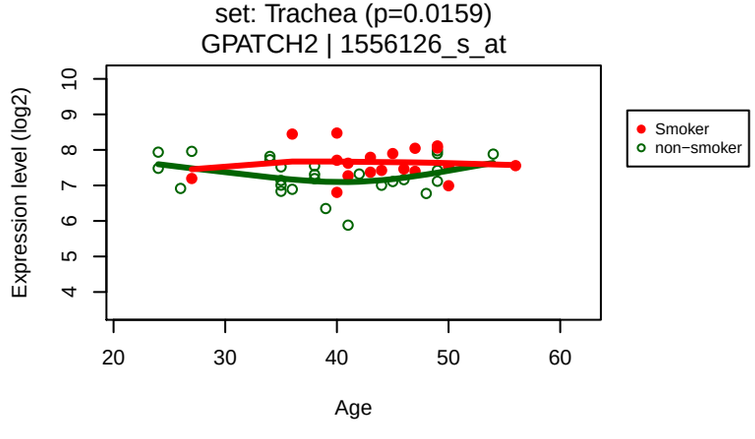

Gene expression ratio in Males
gene: GPATCH2 | 1556126_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0437
p=0.3036
p=0.026
p=0.0064
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPATCH2 | 1556126_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8537
p=0.2009
p=0.4696
p=0.3036
SmallAirway2
LargeAirway2
Trachea
Macrophage2

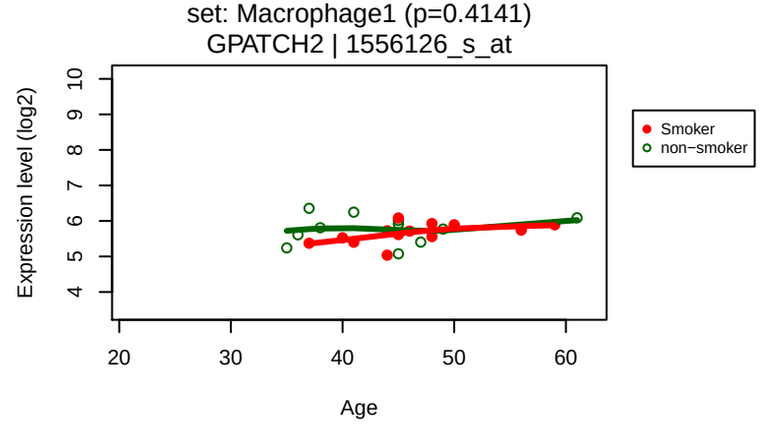

set: SmallAirway1 (p=0.1265)
GPATCH2 | 1556126_s_at
Expression level (log2)
Age
Smoker
non−smoker

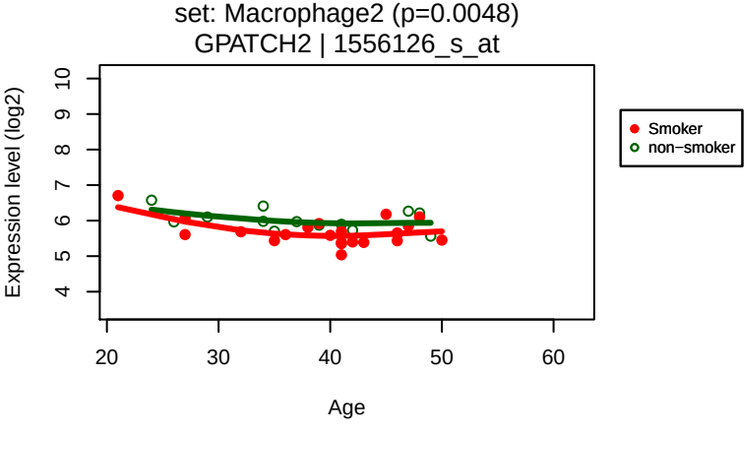

set: SmallAirway2 (p=0.056)
GPATCH2 | 1556126_s_at
Expression level (log2)
Age
Smoker
non−smoker

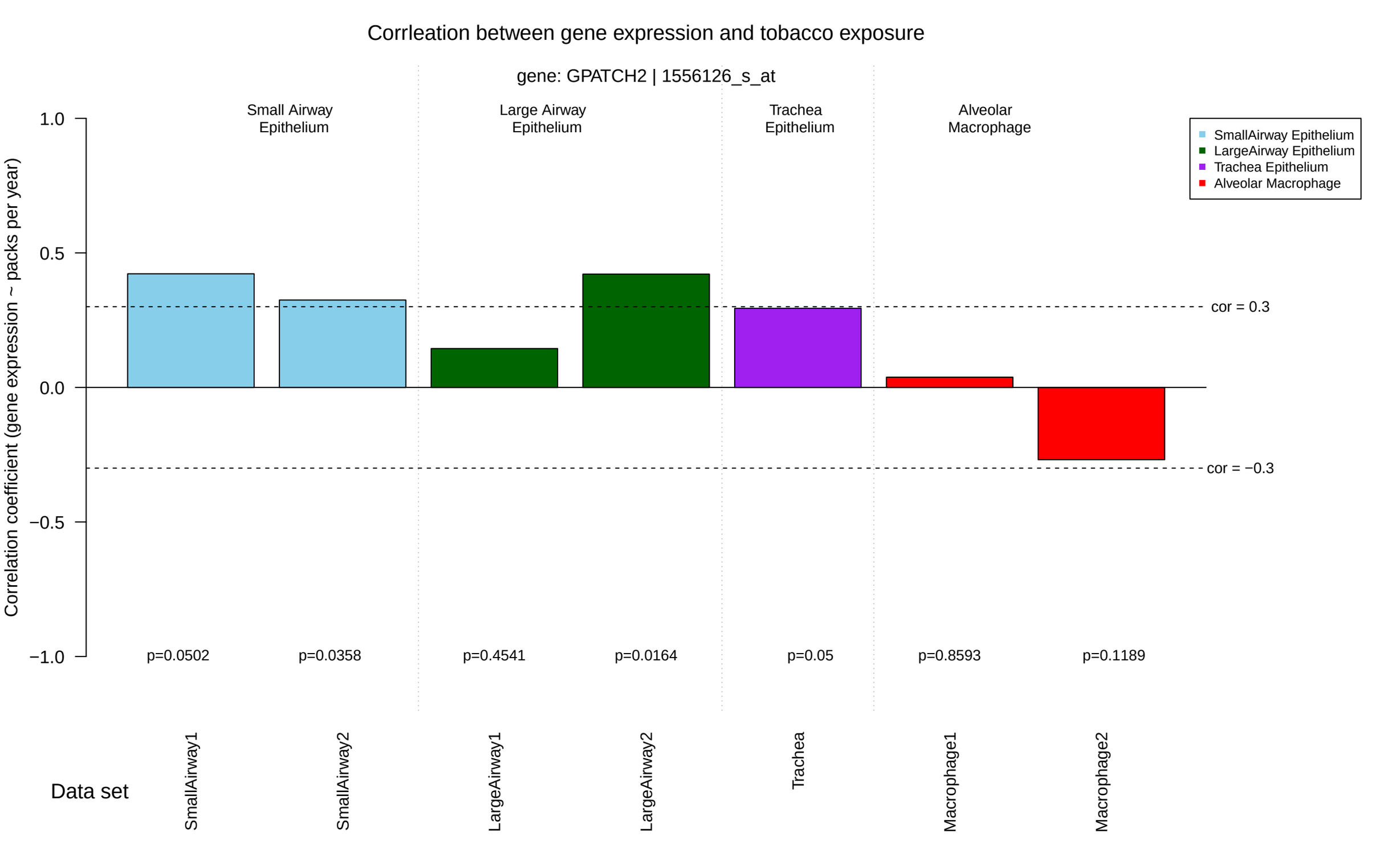

set: LargeAirway1 (p=0.3662)
GPATCH2 | 1556126_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0855)
GPATCH2 | 1556126_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0159)
GPATCH2 | 1556126_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4141)
GPATCH2 | 1556126_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0048)
GPATCH2 | 1556126_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPATCH2 | 1556126_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0502
p=0.0358
p=0.4541
p=0.0164
p=0.05
p=0.8593
p=0.1189
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2